10.Write a Python Program to Implement Expectation & Maximization Algorithm

CODING:

```
import numpy as np

import matplotlib.pyplot as plt

from sklearn.mixture import GaussianMixture

X = np.array([

    [1, 2], [1.5, 1.8], [2, 2.5], [2.2, 2.8],

    [8, 8], [8.5, 8.2], [9, 9], [9.5, 8.8],

    [5, 5], [5.5, 5.2], [6, 5.8], [6.2, 6]

])

model = GaussianMixture(

    n_components=3,

    random_state=42

)

model.fit(X)

labels = model.predict(X)

print("Expectation-Maximization Algorithm")

print("-----------------------------------")

print("Cluster Labels:")

print(labels)

print("\nCluster Means:")

print(model.means_)

print("\nCluster Covariances:")

print(model.covariances_)

probabilities = model.predict_proba(X)

print("\nCluster Probabilities:")

print(probabilities)

plt.figure(figsize=(7, 5))
```

```
plt.scatter(
    X[:, 0],
    X[:, 1],
    c=labels,
    s=80
)
plt.scatter(
    model.means_[:, 0],
    model.means_[:, 1],
    marker='X',
    s=200,
    label='Cluster Centers'
)
plt.xlabel("Feature 1")
plt.ylabel("Feature 2")
plt.title("EM Algorithm - Gaussian Mixture Model")
plt.legend()
plt.show()
```

OUTPUT:

```
Expectation-Maximization Algorithm
---------------------------------
Cluster Labels:
[1 1 1 1 2 2 2 2 0 0 0 0]

Cluster Means:
[[5.675 5.5  ]
 [1.675 2.275]
 [8.75  8.5  ]]

Cluster Covariances:
[[[0.216876 0.1875  ]
  [0.1875   0.170001]]

 [[0.216876 0.154375]
  [0.154375 0.156876]]

 [[0.312501 0.2     ]
  [0.2      0.170001]]]

Cluster Probabilities:
[[1.21831061e-29 1.00000000e+00 3.97520891e-54]
 [2.03122526e-17 1.00000000e+00 1.20069797e-58]
 [1.27991803e-14 1.00000000e+00 6.29482195e-47]
 [1.35588372e-14 1.00000000e+00 5.72892021e-42]
 [6.66073438e-12 3.03846321e-47 1.00000000e+00]
 [5.90514070e-10 1.99932992e-52 9.99999999e-01]
 [1.82457150e-21 2.64868190e-64 1.00000000e+00]
 [2.24042135e-14 1.23285169e-65 1.00000000e+00]
 [1.00000000e+00 3.96746469e-12 1.50883352e-16]
 [1.00000000e+00 2.59006268e-15 1.45136350e-15]
 [1.00000000e+00 1.19818546e-20 7.08434470e-11]
 [9.99999997e-01 1.87015930e-22 3.01025341e-09]]
```